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Bihar Engineering University, Patna

2013 (A)

Time: 3 hours

Code: 103307 ***Subject: Electromagnetic Field Theory*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are TEN questions in this paper.
(iii) Attempt any FIVE questions.

Q.1 Solve both questions : **[14]**

(a) Find the potential distribution due to a long pair of parallel wires of negligible cross-section and having equal and opposite line charge density. Also obtain equipotential surfaces produced by them.

(b) Find the capacitance of two parallel cylindrical conductors having their radii as a and separation between their axes as b.

Q.2 Solve all questions : **[14]**

(a) State uniqueness theorem and prove it.

(b) Explain conductor properties and obtain boundary conditions.

(c) For a two-dimensional system in which $r=\sqrt{(x^2+y^2)}$ determine $\nabla^2 V$ when $V=1/r$.

Q.3 Solve all questions : **[14]**

(a) Find the energy density in the magnetic field.

(b) Find the magnetic field inside a solid conductor carrying a direct current, and hence obtain total magnetic flux per unit length within the conductor.

(c) Prove Stokes' theorem.

Q.4 Solve both questions : **[14]**

(a) Obtain two Maxwell's equations which deviate from steady-state field.

(b) The electric field of electromagnetic wave is given by $E_x=0=E_z$, $E_y=A\cos\omega(t-z/c)$. Using Maxwell's equation in free space, find the magnetic vector H.

Q.5 Solve both questions : **[14]**

(a) Find the ratio of E and H in a uniform plane wave.

(b) Discuss the wave propagation in conducting medium and obtain the value of α and β.

Q.6 Solve this question : **[14]**

Derive the reflection coefficient of perfect dielectric for oblique incidence in the case of parallel polarization. Obtain Brewster angle.

Q.7 Solve both questions : **[14]**

(a) State Poynting theorem and prove it.

(b) A short vertical transmitting antenna erected on the surface of a perfectly conducting earth produces effective field strength $E_{eff.}= E_{\theta eff.}= 100\sin\theta[mu]$ / [m] at points at a distance of one mile from the antenna. Compute the Poynting vector and total power radiated.

Q.8 Solve both questions : **[14]**

(a) Discuss UHF line as circuit element and hence find the input impedance of short-circuited quarter-wave line.

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(b) Discuss quarter-wave line as transformer.

Q.9 Solve both questions : **[14]**

(a) Discuss Smith chart and its uses.

(b) Design a necessary matching unit to join without impedance mismatch the two different sections of transmission line whose impedances are 75 ohm and 50 ohm.

Q.10 Solve this question : **[14]**

Find the field component of TM wave in parallel plane guide and hence discuss TEM wave.

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